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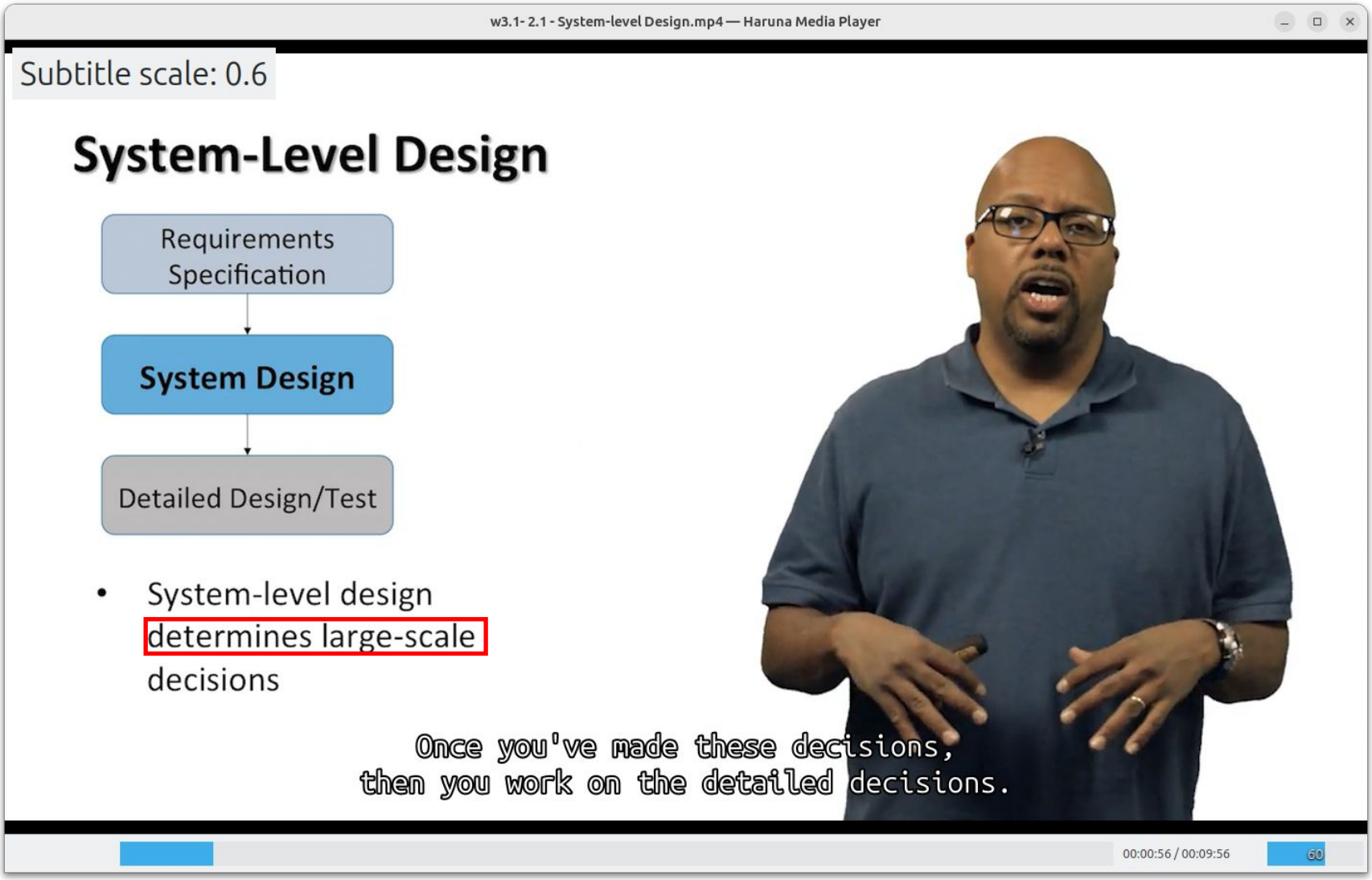

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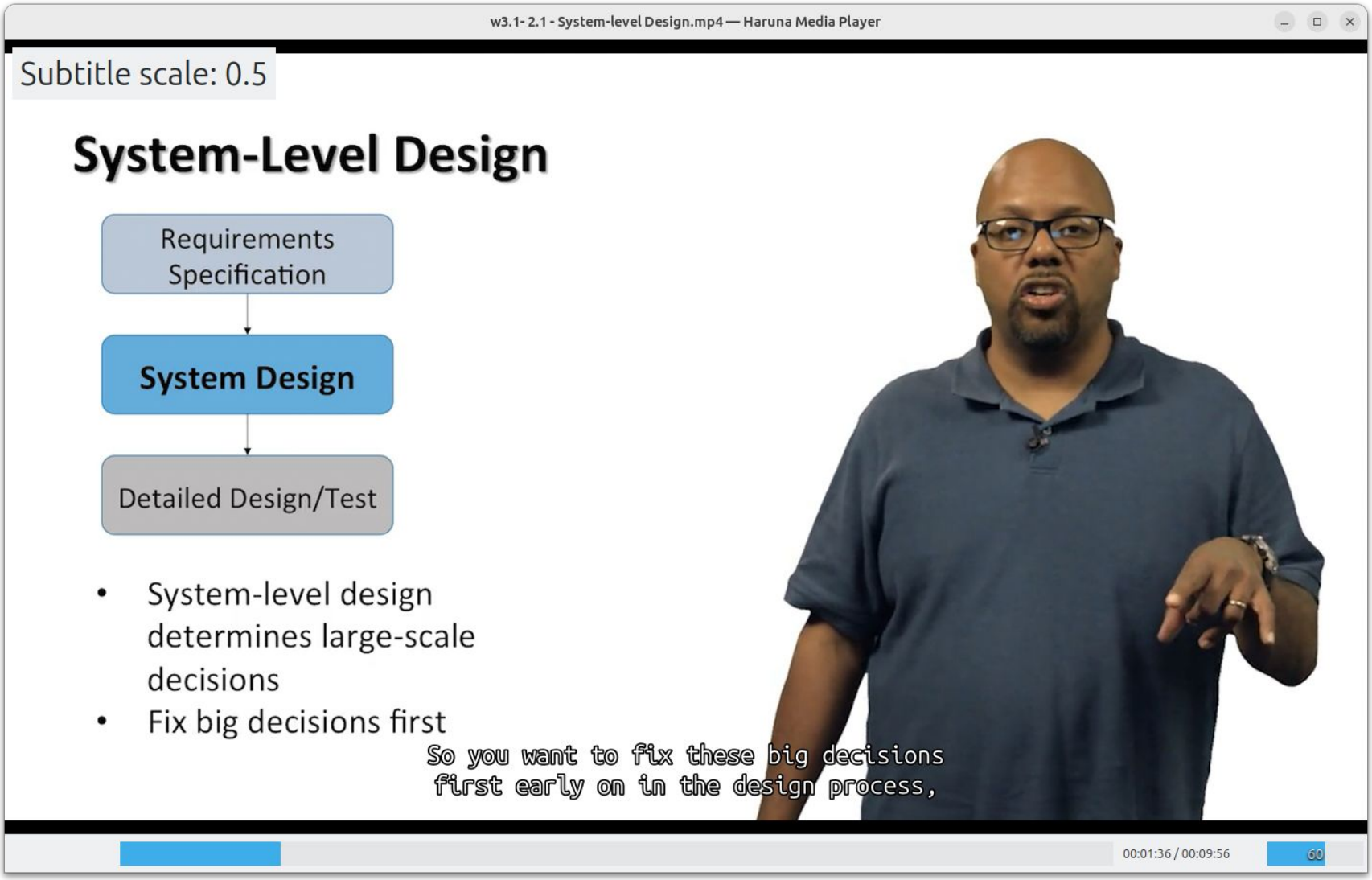

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Designing Within Constraints

- Requirements specification provides constraints to satisfy
- Design **must meet all constraints**
- System-level decisions are those that have a big impact on the constraints
- Making the wrong system decision can make it impossible to meet constraints

speed constraint because you made this
wrong high-level decision, and then what

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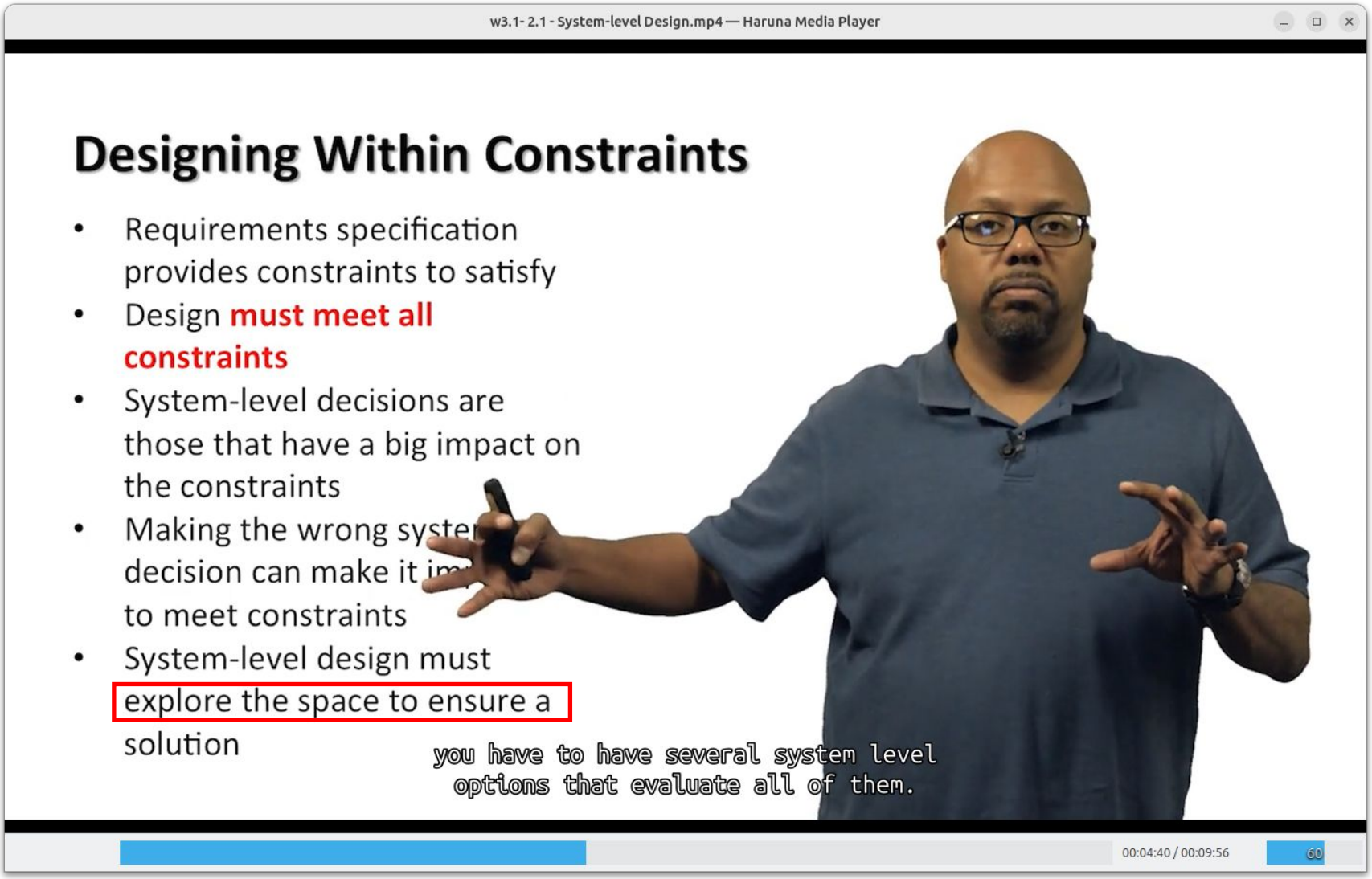

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Design Alternatives

- You must demonstrate that you have explored design alternatives
- **Enumerate design alternatives**
- **Evaluate each alternative**
- Document this process
- Example: Building an RC Quadcopter

1. Buy a kit
2. Build from scratch
3. Use a pre-built flight controller
   - **Estimate** the characteristics of each option

I can use a pre-built flight controller,
but I can build the rest from scratch.

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Summary

1. System Design vs. Detail Design

- **System Design (Large-Scale):** This is where you make the "big" decisions early. These choices determine if you can actually meet your constraints.
- **Detail Design (Small-Scale):** Once the system architecture is fixed, you move into the specifics of each component.
- **The Flow:** While these seem like separate steps, they often overlap. However, for your Capstone, you must clearly document the system-level decisions first.

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2. Designing Within Constraints

Your **Requirement Specification** acts as a boundary. You must satisfy high-level constraints such as:

- **Performance:** Speed, computation rates, or latency.
- **Cost:** Budget limits (e.g., staying under $100).
- **Power:** Battery life and consumption.

> **Ninja Warning:** Making a wrong system-level decision (like choosing an Arduino when you need the high-speed processing of a Raspberry Pi) can make it **impossible** to satisfy your constraints later, wasting time and money.

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3. Design Space Exploration

You are required to investigate multiple **Design Alternatives**. Don't just pick the first idea that comes to mind!

- **Enumerate:** List at least 2 or 3 different ways to build your system.
- **Evaluate & Rule Out:** Use estimation or research to justify why one option is better than another.
- **Research Previous Work:** Look for other Capstone projects, blogs, or forums (like GitHub or engineering sites). See what worked for others and what failed.

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4. Practical Example: The Quadcopter

To illustrate, consider these three system-level alternatives for an RC Quadcopter:

1. **Buy a Kit:** High cost, low effort, guaranteed to work.
2. **Build from Scratch:** Low cost (potentially), high effort, high risk of failure.
3. **Pre-built Flight Controller + Custom Frame:** Middle ground; uses a reliable brain but allows for custom mechanics.

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Packet Sniffer Example

Option1: **Laptop plus sniffer software** (wireshark)

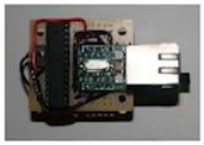
- Cost: $500; Size: Large; Difficulty: Easy

Option 2: **Pwnie Express**

- Small Linux platform
- Cost: $380; Difficulty: Medium

Option 3: **Hardware Packet Sniffer**

- Small and cheap
- Cost: $40; Difficulty: Hard



you can do it in the time required,
things like this.

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Key System Design Decisions

HW/SW= Hardware and Software

1. How will the behavior be divided into HW/SW components?
2. Will components be pre-fab or built from scratch?
3. Do component interfaces match?

- Example: **"Target a Red Box"** system
  - "Red Detection code" and "Servo code" on same microcontroller?
  - Should Red Detection be done using an off-the-shelf component?
  - Can microcontroller interface with off-the-shelf component?

servos and I'm going to move the servos
until that red is dead center in my image,

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1. Hardware/Software Partitioning

The first major decision is determining **behavior division**.

- **Hardware (HW):** Generally faster and more energy-efficient but rigid (cannot be easily changed after manufacturing).
- **Software (SW):** Flexible and easier to update, but often slower and more taxing on the processor.
- **The Goal:** Find the balance where performance requirements are met without over-complicating the physical hardware.

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2. Pre-fab vs. Built from Scratch

This is the "off-the-shelf" vs. "custom" debate.

- **Built from Scratch:** Allows for a highly optimized, proprietary design that fits exact needs. However, it is expensive and time-consuming.
- **Pre-fab / Off-the-shelf:** Faster to market and often cheaper, but you may have to deal with "bloat" (features you don't need) or physical constraints that don't perfectly match your vision.

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3. Interface Matching

Even if you choose the best individual components, the system fails if they cannot "talk" to each other.

- **The Challenge:** Ensuring that the output of one component (like a sensor) matches the input requirements of another (like a microcontroller).
- **Example: "Target a Red Box" System:**
  - If you use an off-the-shelf camera for "Red Detection," does its data format work with your existing servo motor code?
  - Can your microcontroller handle the processing load of an off-the-shelf vision component, or will it crash?

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System Design Decision Example

- Designing a Multicopter
  - Frame: pre-fab or DIY
  - Flight controller: pre-fab or DIY
  - How many motors?
  - Which motors/propellers?
  - Do we need GPS?
  - What type of transmitter/receiver?
- Decisions made based on constraints
- No programming at this stage
- No detailed wiring, but I/O is considered
  - Are enough pins available?

pins in your devices to
communicate with each other, but

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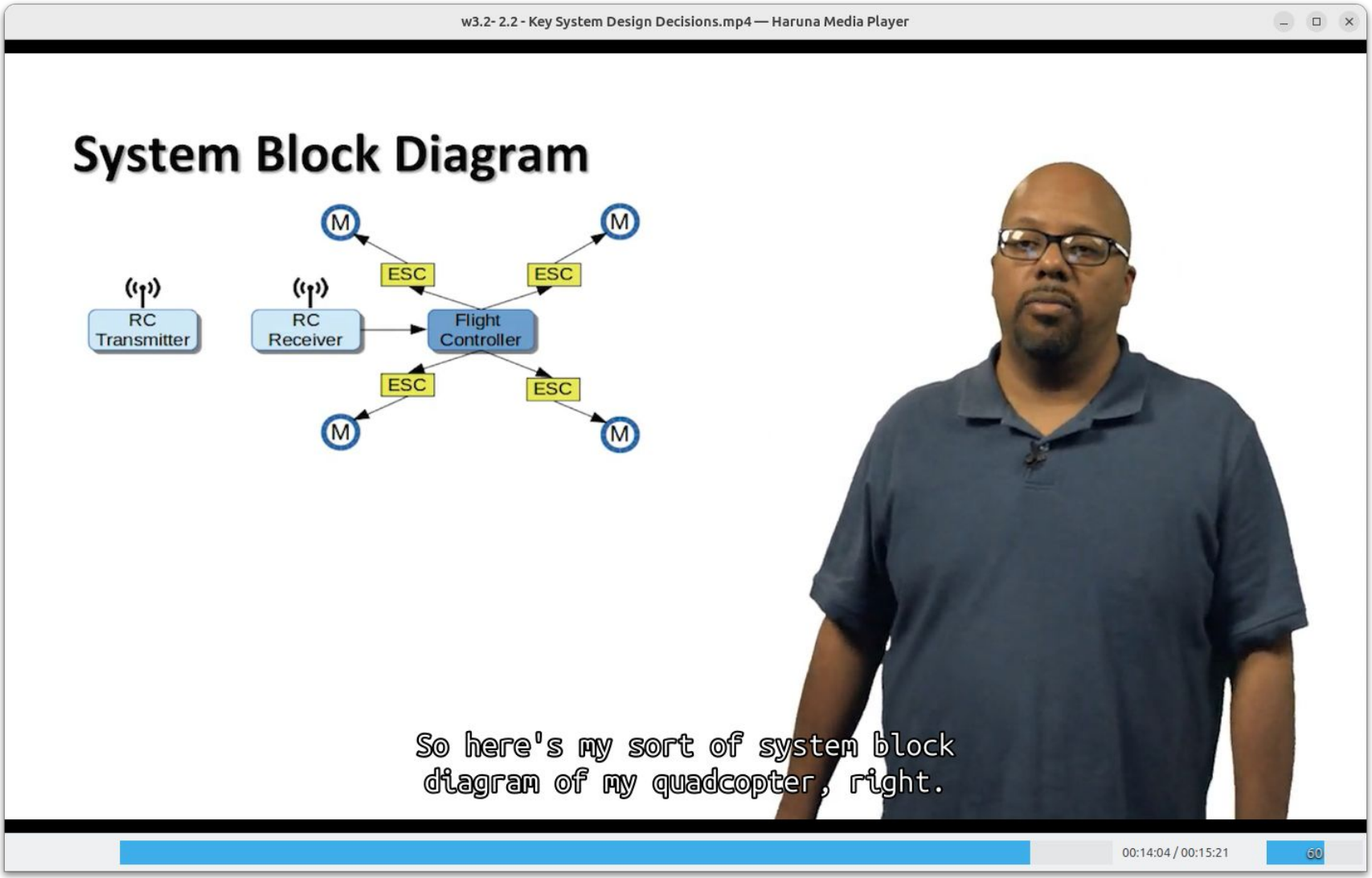

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Summary

This lecture dives into the practical application of **System Level Design Decisions**, using real-world examples to illustrate how to weigh options based on cost, size, and complexity.

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1. Case Study: The Stealth Packet Sniffer

The goal was to create a device for "Penetration Testing"—an authorized attack on a network to find security flaws. The device needed to be small, cheap, and able to exfiltrate (send out) data remotely.

Option	Hardware Choice	Cost	Size
1	**Laptop + Wireshark**	High ($500)	Large
2	**Pwnie Express**	Medium ($380)	Small
3	**Custom Hardware**	Low ($40)	Tiny

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2. Critical System Decisions

When you are moving from your **Requirement Specification** to your design, you must decide:

- **Hardware/Software Partitioning:** Which tasks will software handle (easier to code) and which need dedicated hardware (faster/cheaper but harder to design)?
- **Prefabricated vs. Scratch:** Will you buy a "pre-fab" board (higher cost, saves time) or build it from a raw chip (lower cost, takes months)?
- **Interface Matching:** Do your components speak the same language?
  - *Example:* Does your sensor use **I2C** while your microcontroller only supports **SPI**?
  - *Pin Count:* Do you have enough physical pins on your controller to connect everything?

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3. Example: Target a Red Box (Visual Tracking)

If you want to point a laser at a red object, you have two main system paths:

- **Path A:** Use a standard camera and write your own complex image-processing code on a Raspberry Pi.
- **Path B:** Use an off-the-shelf component like a **PixyCam**. It has its own processor that handles the "red detection" for you and just tells your Arduino the coordinates.

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4. The System Block Diagram

In this phase, you are **not** doing detailed wiring or programming. Instead, you create a **Block Diagram** to show the flow of control and data.

- **Receiver:** Captures your remote signals.
- **Flight Controller:** The "brain" that calculates motor speeds.
- **ESC (Electronic Speed Controller):** Translates the brain's signals into high-power electrical pulses for the motors.
- **Motors/Propellers:** Provide the physical lift.